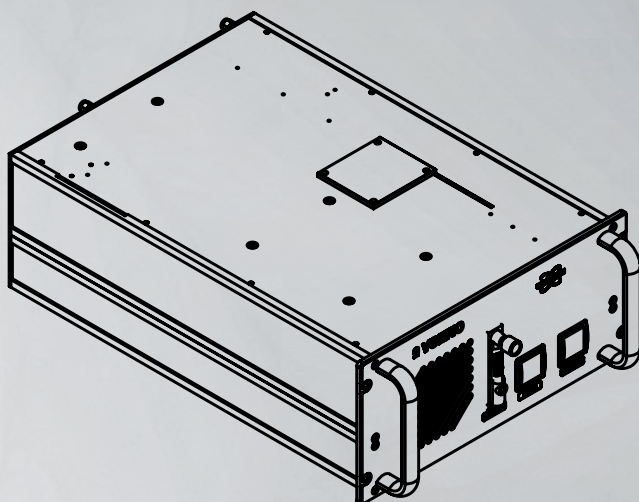


OMEGA 5

- User Manual



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The description given in this manual is based on the technical specifications and the technical design valid at the time of publication. Technical specifications and technical design may be subject to change.

This user manual is the original documentation for the OMEGA 5 gas analyzer.

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1 Introduction

1.1 About this user manual

This user manual contains information on the start-up, operation and maintenance of the OMEGA 5 gas analyzer. Depending on the degree of hazard important safety information and safety instructions are classified in this user manual as follows:



WARNING

- Indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION

- Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

NOTICE

- Indicates a hazardous situation which could result in material damage if the appropriate safety instructions are not observed.



The "i" provides important information on the gas analyzer, or the operation of the gas analyzer.

1.1.1 Terms

This user manual uses both the *gas analyzer* or *OMEGA 5* term when the hardware is described, and the *OPUS GA*¹ term when the software is described, which is used to operate the gas analyzer.

This user manual uses the *OPUS/IR*² term when the spectroscopy software is described, which *OPUS GA* is compatible with.

-
1. Additional OPUS software package for gas analysis
 2. OPUS basic software package

1.1.2 Abbreviations

Table 1.1 contains the most important abbreviations available in the user manual.

Abbreviation	Definition
°C	Celsius
A	Ampere
A/D	Analog/Digital
BaF ₂	Barium fluoride
CaF ₂	Calcium fluoride
ClF ₃	Chlorine fluoride
cm	Centimeter
cm ⁻¹	Wavenumber unit
FT	Fourier Transform
GB	Gigabyte
GHz	Gigahertz
h	Hours
H ₂ SO ₄	Sulphuric acid
HCl	Hydrogen chloride
Hf	Hafnium
hPa	Hectopascal
ID	Inner diameter
IR	Infrared
K	Kelvin
KBr	Potassium bromide
kg	Kilogram
kPa	Kilopascal
m	Meter
mA	Milliampere

Table 1.1: Abbreviations used in the user manual

Abbreviation	Definition
max.	Maximum
mbar	Millibar
MCT	Mercury cadmium telluride
MIR	Mid-infrared
ml	Milliliter
mm	Millimeter
msl	Mean sea level
mW	Milliwatt
NI	Normal flow
nm	Nanometer
OD	Outer diameter
ppb	Parts per billion
ppm	Parts per million
psi	Pound-force per square inch
sec.	Seconds
UPS	Uninterruptible Power Supply
USB	Universal Serial Bus
V	Volt
V AC	Volts alternating current
W	Watt
ZnSe	Zinc selenide

Table 1.1: Abbreviations used in the user manual

1.1.3 Gender-neutral form

The user manual uses the male form in a neutral sense and does not differentiate between male and female users. We would kindly ask all female users to have understanding for this simplified form.

1.2 Safety

Always observe the instructions described in this user manual to ensure user safety and to avoid property damage. Keep this user manual for further reference available at any time.

Improper use or failure to follow the safety instructions can result in serious injuries and/or property damage. Any non-observance infringes the intended use (i.e. spectroscopic measurements) of the gas analyzer. In this case Bruker Optik GmbH does not assume any liability.

It is the operator's duty to plan and implement all necessary safety measures and to supervise their observance. Moreover, the operator must ensure that the gas analyzer is in proper condition and fully functioning. A safe and faultless operation can only be guaranteed if the gas analyzer is transported, stored, installed, operated and maintained properly according to the procedures and instructions described in this user manual.

Never remove or deactivate any supporting safety systems during gas analyzer operation. Objects and/or material not required for operation must not be kept near the gas analyzer operating area.

The gas analyzer has been developed according to the IEC/EN 61010-1 safety regulations for electrical measuring, control and laboratory devices.

1.2.1 Warning labels

When operating the gas analyzer you have to observe a number of safety instructions which are highlighted by the appropriate warning label. The warning labels and their meaning are described in the following.

All warning labels on the gas analyzer must always be kept legible. Immediately replace worn or damaged labels.

Label	Definition
	General Hazard: This warning symbol indicates general hazard. Observe the safety instructions and follow the precautions described. Any non-observance may cause personal injury.
	Hot Surface: This warning symbol refers to components and surfaces which can become very hot during gas analyzer operation. Do not touch these components and surfaces. Risk of skin burn! Be careful when operating near hot components and/or surfaces.

Table 1.2: Warning labels

Label	Definition
	Asphyxiation: This warning symbol indicates gases (e.g. gaseous nitrogen) required to operate the gas analyzer (e.g. purging the gas analyzer). Working in badly ventilated rooms may cause an asphyxiation risk. Always make sure that the rooms are well ventilated.

Table 1.2: Warning labels

1.2.1.1 In case of hazardous sample material

Besides the dangers described above, there can also be hazards caused by the sample material. Depending on the type of hazardous substances used, you have to observe the specific substance-relevant safety instructions. Put on the specific warning label on the corresponding module position. The label must be legible and permanently discernible.

The following list exemplifies types of hazardous sample material:

Label	Definition
	Infectious Material: This warning symbol indicates the possible existence of biologically dangerous and infectious material. When working with this kind of material always observe the prevailing laboratory safety regulations and take necessary precautions and disinfection measures (e.g. wearing protective clothing, masks, gloves etc.). Non-observance may cause severe personal injury or even death. For information on how to use, dilute and efficiently apply disinfectants, refer to the Laboratory Biosafety Manual: 2004 by WHO - World Health Organization.
	Radioactive Material: This warning symbol indicates the possible existence of radioactivity. When working with radioactive material always observe the safety regulations and take necessary protective measures. Wear protective clothing, e.g. masks and gloves. Non-observance may cause severe personal injury or even death.
	Corrosive Substances: This warning symbol indicates the possible existence of corrosive substances. When working with corrosive substances always observe the laboratory safety regulations, and take protective measures (e.g. wear protective masks and gloves). Non-observance may cause severe personal injury or even death.

Table 1.3: Warning labels in case of hazardous sample material

1.2.2 Waste disposal

Dispose all waste produced (chemicals, infectious and radioactively contaminated substances etc.) according to the prevailing laboratory regulations. Detergents and cleaning agents must be disposed according to the special waste regulations.

1.3 General information

1.3.1 Protective earthing

To avoid personal injuries and/or damage caused by electrical power, the supplied gas analyzer power cord is equipped with a safety plug. Connect this safety plug only to an earthed power socket. Make sure that the earthed power socket used complies with IEC (International Electrotechnical Commission¹).

1.3.2 Qualified personnel

Initial installation and all maintenance and repair work not described in this manual should only be performed by Bruker Service. Make sure that the gas analyzer is only operated and maintained (i.e. only maintenance work that is described in this manual) by authorized operating personnel trained in the gas analyzer operation and all relevant safety aspects.

All repairs, adjustments and alignments on any gas analyzer component must be performed in accordance with the safety regulations and standards applied in the country where the instrument will be installed.

1.3.3 Intended use

OMEGA 5 is an FTIR gas analyzer for spectroscopic analysis of gas phase concentrations in the mid-infrared range. The gas analyzer is suited for the measurement of gas concentrations in non-flammable gases and vapors, i.e. gases or vapors with a flammable content which, under any circumstance, is always below the lower explosive limit. In terms of the particular application, the operator is obliged to monitor and comply with the aforementioned measuring conditions with regard to the gases and/or vapors used.

The intended use includes applications like e.g.:

- analysis of industrial exhaust gases
- determination of gas impurities
- process surveillance in production lines
- investigation of catalytic processes

1. International standards organization for electrical and electronic-related technologies.

The gas analyzer is intended for indoor use only, under the environmental conditions specified in appendix A.6.

The intended use of the gas analyzer complies with the relevant standards and regulations, especially:

- national and local safety regulations
- national and local accident prevention regulations
- generally recognized technical regulations

The operator has to strictly observe all instructions given in this user manual, i.e.:

- safety instructions
- installation instructions including the installation site requirements
- operation instructions
- repair and maintenance instructions

Any other use of the gas analyzer not described in this user manual is not of intended use, and can lead to personal injuries and/or property damage.

WARNING



Gas analyzer is not operated in accordance with its intended use

Risk of death or serious injury

- Do not take any actions or measures that violate the intended use of the gas analyzer to ensure the personal safety of the operator and the operational safety of the gas analyzer.

The gas analyzer may only be operated in connection with the system components and instructions described in this manual, and according to the instructions advised by Bruker Service.

In case of accessories or components made by other manufacturers and used in connection with the gas analyzer, Bruker Optik GmbH does not assume any liability for safe operation and proper functioning.

1.4 Questions and concerns

If you have questions or concerns about safety, operating the gas analyzer, or if you need assistance with software problems or replacement parts, contact Bruker at the numbers listed below:

- Service hotline hardware: +49 (0) 72 43 504-2020
- Service hotline software: +49 (0) 72 43 504-2030
- Fax: +49 (0) 72 43 504-2100
- E-mail: service.bopt.de@bruker.com
service.bopt.us@bruker.com
- Internet: www.bruker.com/optics
- International service: www.bruker.com/about-us/offices/offices/bruker-optics

2 Overview

The OMEGA 5 gas analyzer is a compact FTIR gas analysis system, installed in a 19" rack-mount box, and equipped with a 5 m multi-reflection gas cell for the measurement of industrial gases.

2.1 Gas analyzer configuration



OMEGA 5
FTIR gas analysis system

The gas analyzer allows a precise and automated monitoring of gas concentrations in real time. A calibration with regard to the analysis gas is not required.

2.1.1 Optional configuration

In the standard gas analyzer configuration, the gas cell is not suited to be operated using overpressure above ± 1.0 bar (± 14.5 psi). The high-pressure version OMEGA 5 HP allows to operate the gas cell at an overpressure of 14.0 bar (203.0 psi).

2.2 Assembly

The gas analyzer is equipped with the following hardware components:

- multi-reflection gas cell, 5 m
- RockSolid Interferometer
- thermoelectrically cooled MCT detector
- CenterGlow IR source

Due to its modular design the maintenance tasks can be performed without any re-alignment of the optics. Details on maintenance are described in chapter 6, on spare parts and/or consumables in appendix B.

2.3 Applications

The gas analyzer can be used for the following application fields:

- process monitoring in production lines
- exhaust gas tests of smokestacks
- motor vehicle exhaust analysis
- biogas analysis
- determination of gas impurities
- scientific research

A quantitative and qualitative evaluation of the measuring results can be performed using the OPUS GA software.

2.4 Optics

The gas analyzer incorporates state-of-the-art optical components for outstanding sensitivity and stability, which enable:

- high quality measurement results
- less down-time

In the standard configuration, the gas analyzer is equipped with a thermoelectrically cooled MCT detector. The CenterGlow IR source optimizes the location of the glowing area within the source and maximizes the light flux.

2.5 Electronics

The gas analyzer electronics are based on a high speed 24 bit data sampling unit that guarantees experimental results with an outstanding accuracy. Any modern data system (PC workstation, laptop etc.) with the OPUS/IR spectroscopy software and OPUS GA analysis software installed can be used to control the gas analyzer and perform data processing.

The gas analyzer is connected to the data system by a standard 100Base-T Ethernet connection, which allows the gas analyzer to be integrated into an existing data network. Details on the electronics specifications are described in appendix A.

2.6 Software

The OPUS GA (OPUS gas analysis) software package evaluates the measured spectra in real-time to identify and quantify gas compounds.

The quantification is based on a non-linear fitting algorithm that fits the corresponding library spectrum to the measurement. This fitting procedure also includes the absorptions of interfering gases.

More than 300 gas compounds can be identified and quantified without the need of a calibration measurement.

3 Installation

Unpacking and initial installation of the gas analyzer is done by Bruker Service. The operating company has to provide an installation site that meets the site requirements described in this chapter.

3.1 Packaging

NOTICE

Damaged packaging

Property damage

- Contact shipping company.
- Do NOT put the gas analyzer into operation.

3.2 Scope of delivery

The following components are part of the scope of delivery for the basic configuration of the gas analyzer:

	Piece	Definition
1	1	OMEGA 5 gas analyzer with 5 m gas cell, integrated temperature controllers for the gas cell and transfer line, and a purge flow regulator to regulate the purge gas
2	1	Country-specific power plug
3	1	Data cable (100Base-T cross-over cable)
4	1	OMEGA 5 user manual
5	1	OPUS/IR basic spectroscopy software package

Table 3.1: Scope of delivery for the basic spectrometer configuration

3.3 Transportation

Due to its weight (28 kg) the gas analyzer has to be carried by at least 2 persons to avoid personal injury.

For service and maintenance purposes, always transport the gas analyzer in the original packaging to avoid property damages on the gas analyzer due to improper packaging.

3.4 Site requirements

The operating company has to provide an installation site that meets particular requirements. More details are described in appendix A.

Installation site:	• In a closed room, up to a maximum height of 2000 m above sea level • Room atmosphere must not be explosive • Do not expose to strong drafts, direct sunlight or high air humidity • The installation site must be free of any vibration sources, e.g. ventilation hoods, air conditioners, motors, floor vibrations etc.
Space:	• 740 x 485 x 222 cm (d x w x h) • Stable and horizontal base, which can withstand the weight of about 30 kg • For proper ventilation, keep the venting slots free at the gas analyzer rear side • Keep a distance of about 15 cm between the wall and the gas analyzer rear side • Distance between data system and gas analyzer must not exceed 30 m, when directly connected

Table 3.2: Site requirements

Environment:

- | | |
|----------------|---|
| Temperature: | <ul style="list-style-type: none"> • Operating temperature range: 5°C to 35°C • Storage and transport temperature: -20°C to 70°C • Humidity (non-condensing): less than 80% (relative humidity) • Temperature variations: less than 1°C/hour and not more than 2°C/day in case of long-term measurements |
| Interferences: | <ul style="list-style-type: none"> • Do not install devices such as large electric motors, heaters, welding equipment, radio transmitting equipment, units emitting pulsed NMRs, or high-powered lasers in close vicinity to the gas analyzer. These devices can cause a gas analyzer malfunction. • Make sure that these types of devices are not connected to the same electrical circuit as the gas analyzer. • Do not install the gas analyzer near any sources of potential inductive electrical interference (e.g. pumps, switching motors, microwave ovens etc.), sources of high energy pulses, and sources that might cause magnetic or radio frequency interference. |

Table 3.2: Site requirements

Power supply:	
Operating voltage:	110 V ±10%/60Hz or 230 V ±10%/50Hz Use an UPS (<u>U</u>ninterruptible <u>P</u>ower <u>S</u>upply) unit to ensure an uninterruptible power supply, and subsequent uninterruptible operation, if there are any problems concerning main power supply caused by e.g. drop-outs and flickers, power surges, frequent thunderstorms etc.
Operating current:	• 8 A, in case of 110 V • 4.5 A, in case of 230 V
Protection class:	I, according to IEC 61140
Main power supply:	The main power supply must be easily accessible at any time, and easily interruptible if required by either unplug the power cable from the mains outlet, or from the external power supply unit.
Line-powered accessories:	• Line-powered accessories connected to the spectrometer interfaces such as Ethernet have to provide special electrical disconnecting features. • The electric circuits of these interfaces have to comply with SELV circuit (Safety Extra Low Voltage circuit) requirements, even if the connection has already been realized. i Typically, this is achieved when connecting SELV circuits to each other. In general, the interface requirements are met if the accessories comply with the regulations outlined in EN 61010 (Safety regulations for laboratory equipment) or EN 60950 (safety for information technology facilities).

Table 3.2: Site requirements

3.5 Connection ports

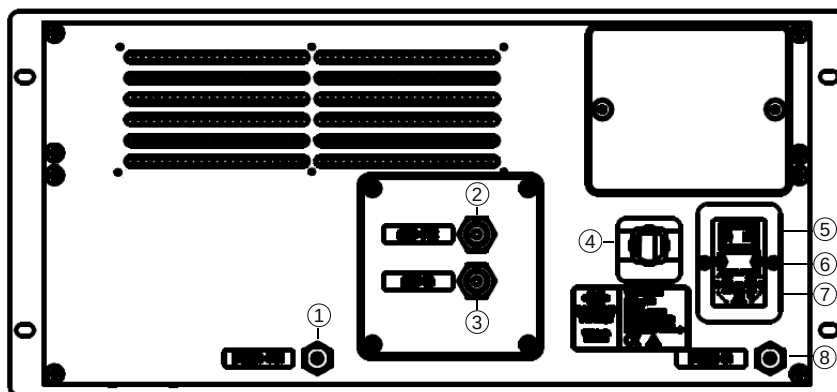


Figure 3.1: Gas analyzer rear side with ports

	Port	Definition
1	Purge gas out	The port allows to transfer purge gas out of the gas analyzer. ➤ Required: connecting hose, OD 6 mm^a Details on the purge gas requirements are described in chapter 5.2.2.
2	Gas out	Swagelok for hose/tube with an outer diameter of 6 mm
3	Gas in	Swagelok for hose/tube with an outer diameter of 6 mm
4	Ethernet port	Port to connect a computer or printer The port is designed for RJ-45 plugs and complies with the 10/100Base-T Ethernet standard.
5	Power switch	I/O switch (ON = I, OFF = O)
6	Primary power receptacle	Receptacle to connect the power cord
7	Fuse holder	The fuse holder contains the following fuse types: • 2 x T 8 AH, in case of 110 V • 2 x T 4 AH, in case of 230 V Details on how to replace the fuses are described in chapter 6.2.

Table 3.3: Definition of ports located at the gas analyzer rear side

	Port	Definition
8	Purge gas in	The port allows to transfer purge gas into the gas analyzer. ➤ Required: connecting hose, OD 6 mm^a Details on the purge gas requirements are described in chapter 5.2.2.

Table 3.3: Definition of ports located at the gas analyzer rear side

a. The connecting hose can be ordered at Bruker. Order number: 11619 (PNM HOSE BK POLYETHYLENE DA=6)

3.6 Installing hardware

For initial installation, all necessary hardware components are connected and installed by Bruker Service. The gas analyzer can be installed in a 19" rack, or used as a desktop unit.

When relocating the gas analyzer, observe the installation steps for the standard configuration described in chapter 3.6.1 et seq. In case of questions on hardware installation contact Bruker Service. The contact details are described in chapter 1.4.

3.6.1 Connecting gas analyzer to power supply

1		Put the I/O switch in the "O" (OFF) position. ➤ The I/O switch is located at the gas analyzer rear side.
2		Connect one end of the power cord^a to the power port located at the gas analyzer rear side.

Table 3.4: Connecting gas analyzer to power supply external power supply unit

3	Connect the power plug of the power cord to the mains socket outlet.  CAUTION Power plug of the power cord connected to a socket outlet missing earthing contact. Risk of personnel injury ➤ ALWAYS connect power plug to a socket with earthing contact. NOTICE Power plug of the power cord connected to a socket outlet missing earthing contact. Property damage ➤ ALWAYS connect power plug to a socket with earthing contact.
---	--

Table 3.4: Connecting gas analyzer to power supply external power supply unit

- a. Depending on the local requirements and regulations, the original power cord may need to be replaced by a power cord commonly used within the particular country. The cord must be technically approved by the local authority, e.g. UL for USA, CSA for Canada or VDE for Europe. An original power cord with a country-specific connecting plug can be ordered with Bruker.

3.6.2 Connecting gas analyzer to purge gas supply

i The connecting hose (OD: 6 mm) required is not part of the standard delivery content and must be provided by the operator with the correct length. The connecting hose used must be suited for the operating pressure and operating temperature.

1. At the gas analyzer rear side, rotate the Swagelok¹ standard plugs of the purge gas inlet and outlet counterclockwise, and remove the plugs.
2. Mount - but not tighten - the Swagelok nut, including the ferrule set onto the open gas inlet.
3. Insert the connecting hose into the purge gas inlet (② figure 3.2).

1. Manufacturer name of component, www.swagelok.com

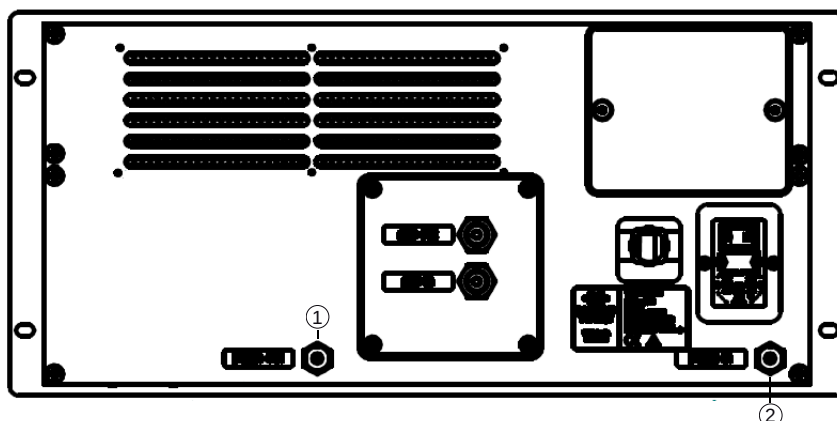


Figure 3.2: Purge gas inlet and outlet

4. Tighten the Swagelok nut clockwise until the connecting hose cannot be rotated manually anymore, or move axially in the fitting.
➤ *For more details on using Swagelok tube fittings, observe the manufacturer assembly instructions.*
5. Connect the other end of the connecting hose to the purge gas supply on site.
6. To equalize the pressure, keep the purge gas outlet (① in figure 3.2) open to allow the gas to evaporate.

Details on purging and purge gas requirements are described in chapter 5.2 et seq.

3.6.3 Connecting gas analyzer to analysis gas supply

i The connecting hose (OD: 6 mm) or tube (hereinafter referred to as *connecting hose*) required is not part of the standard delivery content and must be provided by the operator with the correct length. The connecting hose used must be suited for the operating pressure and operating temperature as well as for the analysis gas.

1. At the gas analyzer rear side, rotate the Swagelok¹ standard plugs of the (target) gas inlet (① in figure 3.3) and gas outlet (②) counterclockwise, and remove the plugs.

1. Manufacturer name of component, www.swagelok.com

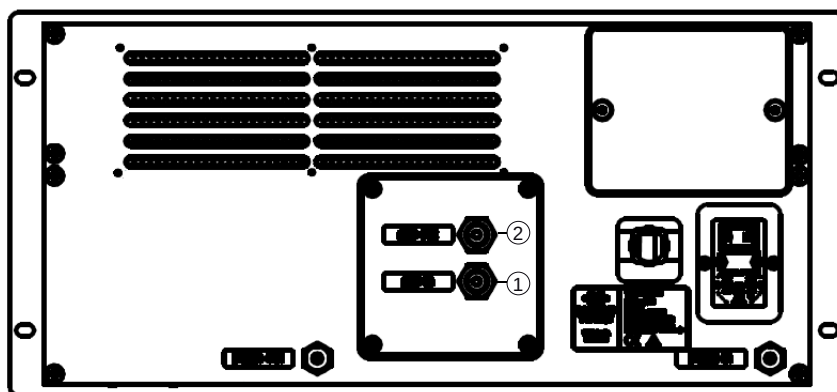


Figure 3.3: Gas inlet and gas outlet

2. Mount - but not tighten - the Swagelok nut, including the ferrule set onto the open gas inlet.
3. Insert the connecting hose into the (target) gas inlet (①).
4. Tighten the Swagelok nut clockwise until the connecting hose cannot be rotated manually anymore, or move axially in the fitting.
5. Connect the other end of the connecting hose to the analysis gas supply on site.

Details on the analysis gas requirements are described in appendix A.9.

3.6.4 Connecting gas analyzer to a PC

The gas analyzer and the data system are already pre-configured for stand-alone operation by Bruker. For connecting the gas analyzer to a stand-alone PC use the standard data cable supplied. It is a crossover cable for 100Base-T Ethernet standard and RJ45 plugs.

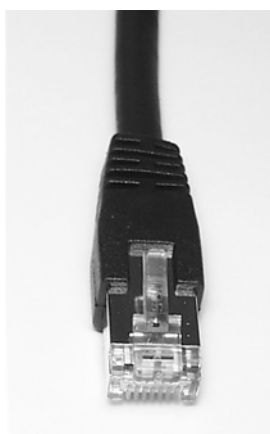


Figure 3.4: CAT-5 data cable with RJ-45 plug

The data cable supplied is determined to be used to directly connect the gas analyzer to a stand-alone PC. Other connection topologies are possible which require, however, a different type of data cable.

Procedure

1. Turn the main power switch into the "O" (OFF) position.
2. Connect one end of the cross-over data cable to the ETH port located on the rear gas analyzer side.
3. Connect the other end of the cross-over data cable to the corresponding PC port. For detailed information refer to the computer manual.

IP address:

- Gas analyzer IP address: preset by Bruker to 10.10.0.1.
- PC IP address: use the same subnet as the gas analyzer, e.g. 10.10.0.x, with x being any number between 2 and 255

3.7 Setting target temperature for integrated transfer line

-  On the left controller, press the  or  key to set the target temperature.
- *The integrated transfer line starts heating to reach the target temperature set.*

3.8 Setting target temperature for gas cell

-  On the right temperature controller, press the  or  key to set the target temperature.
- *The gas cell starts heating to reach the target temperature set.*

3.9 Gas analyzer operability

After switching on the gas analyzer, there is a beep if the connection to the PC has been established.

3.10 Spectroscopy software

3.10.1 Gas analyzer configuration with Bruker PC or notebook

The OMEGA 5 gas analyzer is supplied by default with the spectroscopy software OPUS/IR developed by Bruker. Optionally, the OPUS GA software can be ordered, which has specifically been designed for the application in gas analysis.

If the gas analyzer is ordered with a PC or notebook supplied by Bruker, the supplied spectroscopy software is pre-installed on this device.

3.10.2 Gas analyzer configuration with customer PC or notebook

In case of the gas analyzer configuration with the PC or notebook provided by the customer, the OPUS GA spectroscopy software has to be installed on this PC or notebook. The PC or notebook has to meet special system requirements (chapter 3.10.2.1).

The OPUS GA spectroscopy software is provided via a download link, after Bruker Service has created a computer-specific license file.

3.10.2.1 Minimum system requirements of the PC provided by the customer

	PC	Notebook
Processor:	Intel Core 2 Duo, >3 GHz	Intel Core i5, 2.4 GHz
RAM:	4 GB	4 GB
Hard disk:	500 GB HDU or better	250 GB HDU or better
Operating system:	Windows 10	Windows 10
Interfaces:	RJ-45 Network	RJ-45 Network
DVD:	8xDVD	DVD
Flash drive port (2.0 or 3.0):	Yes	Yes

Table 3.5: Minimum system requirements of the PC or notebook provided by the customer

3.10.2.2 Further items to be observed:

- The driver software required for the connected peripheral units has to be installed.
- For hard copies, a color printer is recommended.
- If the data system is to be connected to a network (network interfacing is not part of the spectrometer installation), an additional Ethernet interface is required.

3.10.2.3 Before starting the installation

1. The user must have administrator rights in Windows.
2. The internet explorer must be installed on the PC.
3. The PC must be connected to the spectrometer using the data cable delivered.

3.10.2.4 Installing software

1. Download the zipped *OPUS GA.zip* folder using the provided download link.
2. Unzip the OPUS GA folder and copy it to the C:\ directory of the measurement PC or notebook.

3.10.2.5 After the installation

1. Double click the *OPUS GA.exe* file in the C:\OPUS GA\ folder.
2. Wait a few seconds until the software user interface is shown on the monitor.

4 Hardware Components

This chapter describes all relevant gas analyzer components. Depending on the gas analyzer configuration ordered your instrument may look slightly different.

i The local indications right and left assume that the operator stands in front of the gas analyzer. The same applies to the indication front and rear side, respectively.

4.1 Gas analyzer front side



Figure 4.1: Gas analyzer front side

	Component	Definition
1	Temperature controller for integrated transfer line	The temperature controller allows to set and check the temperature of the integrated transfer line. Details on the temperature controller are described in chapter 4.4.
2	Temperature controller for gas cell	The temperature controller allows to set and check the temperature of the gas cell. Details on the temperature controller are described in chapter 4.4.
3	Flow meter	The flow meter adjusting knob allows to control the purge gas flow rate (permissible flow volumes: 0.4 to 5 NI/min).
4	Venting slots	Cool air is drawn in through the ventilation slots. The ventilation slots must not be blocked.

Table 4.1: Definition of components

4.2 Optics

The gas analyzer optics consists of several components, is sealed and desiccated. The permanently aligned RockSolid interferometer and the diode laser allow precise data acquisition.

4.2.1 IR source

The gas analyzer is equipped with a CenterGlow IR source, which ensures high emission.

The influence of the IR source on the single component values is very large. If the source energy deteriorates, the source has to be replaced by Bruker Service (chapter 1.4).

4.2.2 Detector

By default, the gas analyzer is equipped with a thermoelectrically cooled MCT detector.

4.2.3 Gas cell

The gas cell has been developed to perform absorption measurements of gaseous chemical substances. Alternatively, a spectrum of the gas cell without any sample can be measured and saved as reference spectrum, which has the advantage that the extinction (sum of reflection, absorption, scattering losses of IR light) of the gas cell windows is additionally included in the spectrum. The true transmittance of the sample gas can be calculated from the ratio of the gas cell spectra, with and without sample gas.

CAUTION



Highly toxic, corrosive and/or explosive liquids in connection with the gas cell

Risk of personal injury

- NEVER use highly toxic, corrosive and/or explosive liquids in connection with the gas cell.
- Highly toxic, corrosive and/or explosive liquids in connection with the gas analyzer may damage the gas cell windows due to overpressure or corrosion. The explosive gas mixture may enter the optical compartment of the gas analyzer and detonate due to the hot IR source.

NOTICE**Highly corrosive gas compounds in connection with the gas cell during continuous operation**

Risk of property damage

- Avoid a continuous exposure of the gas cell to highly corrosive gas compounds, e.g. F_2 , C_2 , HF , HCl , H_2SO_4 , ClF_3 , O_3 , at a volume mixing ratio of 10,000 ppm or higher, especially in the presence of water vapor at volume mixing ratios of 1,000 ppm or higher, as the service life of the gas analyzer will significantly be shortened.

The gas cell body consists of an electro-polished nickel-coated aluminum tube. The optical path length of the multi-reflection gas cell is 5 m.

The optics compartment of the gas analyzer is sealed against the gas cell by two windows made of zinc selenide. The standard size of the zinc selenide windows is 19 mm x 4 mm (diameter x thickness). The windows are wedged (6 mrad) and contain a broadband anti-reflection (BBAR) coating by default.

Both the integrated transfer lines at the inlet and outlet of the gas cell, as well as the gas cell itself can be heated up to 191°C to avoid condensation of the gas. The temperature of the transfer line and the gas cell is regulated by separate temperature controllers. The current temperature and pressure inside the gas cell are continuously measured and can be read out by the OPUS GA software.

The optional OMEGA 5 HP gas analyzer configuration contains a 5 m multi-reflection gas cell that can be used for absolute pressures up to 15 bar at temperatures up to 30°C.

This kind of gas analyzer configuration is especially suited to detect very low gas concentrations, such as impurities in gas products.

Applicable volume fractions at atmospheric pressure (1013 mbar) and laboratory temperature (25°C) are in the range of sub-ppm to %. Typical detection limits are of the order of hundreds of ppb to ppm.



The nickel-coated body and sealings of the gas cell are only temporarily (minutes to several hours) resistant against corrosive gases like nitric acid, hydrogen chloride, hydrogen fluoride or ammonia. Never fill corrosive liquids or corrosive solutions into the gas cell.

4.3 Integrated transfer line

The integrated transfer lines transport sample gas to and from the gas cell. Both the integrated transfer lines and the gas cell can be heated to prevent single constituents of the sample gas from condensing.

4.4 Temperature controller

i Before using the temperature controller, carefully read the attached manual for the temperature controller. **IMPORTANT:** if there has already been a setting for the target temperature when the temperature controller is switched on, heating starts immediately after switching on, unless you press the *Esc* and  button simultaneously.

NOTICE

Temperature controller defective

Property damage on gas analyzer
➤ Contact Bruker Service (chapter 1.4).

The temperature controllers for the transfer line (① in figure) and gas cell (②) are located at the gas analyzer front side.



Figure 4.2: Temperature controllers

	Definition
1	Temperature controller for the integrated transfer line ^a
2	Temperature controller for the gas cell ^b
3	Process value (measured temperature)
4	Set point (required temperature)

Table 4.2: Buttons on temperature controller

	Definition
5	Setting buttons:  : home screen of the controller^c  : scroll to select a new parameter  : decreases the target temperature of the integrated transfer line or gas cell^d  : increases the target temperature of the integrated transfer line or gas cell, to a maximum of 191°C^e

Table 4.2: Buttons on temperature controller

- a. If *OP1* flashes on the display, the integrated transfer line is heating.
- b. If *OP1* flashes on the display, the gas cell is heating.
- c. Depending on the application used, you may define the temperature. Further parameters can be set which are described in the user manual of the Eurotherm 3216 temperature controller (<https://www.eurotherm.com/?wpdmdl=26879>).
- d. To decrease the target temperature press *D*. The new set point is entered when you release the button, and indicated by a short flash on the display.
- e. To increase the target temperature press *I*. The new set point is entered when you release the button, and indicated by a short flash on the display.

4.5 FTIR spectrometer

An FTIR spectrometer is integrated into the gas analyzer. The optics of the FTIR spectrometer is sealed and can be purged. The permanently aligned RockSolid interferometer and the diode laser guarantee an accurate and precise data acquisition.

Due to the CenterGlow IR source technology and the thermoelectrically cooled MCT detector installed, OMEGA 5 provides a constantly high performance for many years, independent of ambient temperature.

The need for maintenance is minimized due to the long lifetime of the relevant hardware components like interferometer, IR source and laser.

Details on the FTIR spectrometer are described in appendix A.

4.6 Technical details on evacuation

Evacuating means pumping gases out of a closed container. In most applications, it is about evacuating ambient air.

If no zero air is available to purge the gas cell, in some applications the gas cell can be evacuated to measure a background spectrum. The vacuum level required depends on the type of application. In the end, the quality of the vacuum inside the gas cell during background measurement determines the accuracy of the quantitative gas analysis.

If you use vacuum pumps, make sure that they do not allow any kind of back-diffusion of possible auxiliary agents, e.g. oil from rotary slide-valve pumps. The pump size and performance must be adapted to the gas cell volume. Always observe the specifications of the vacuum pump, and the instructions given in the relevant literature specialized on vacuum technology.

Seal tightness:	Observe the instructions given in connection with the valves used
Vacuum in IR spectroscopy:	Ultimate vacuum within the fine vacuum range is sufficient in most cases
Vacuum pump:	Use low-vibration pumps
Vacuum tubing (corrugated bellow, plastic, rubber):	Put on between pump and gas cell, must be free from angle strain; heavy/long tubing must be mechanically supported
Valves	Carefully screw onto the gas cell inlet

4.7 Optional gas analyzer configurations

4.7.1 OMEGA 5 HP

The optional OMEGA 5 HP gas analyzer configuration contains a 5 m multi-reflection gas cell that can be used for absolute pressures up to 15 bar at temperatures up to 30°C.

This kind of gas analyzer configuration is especially suited to detect very low gas concentrations, such as impurities in gas products.

Feature:	High pressure
Temperature sensor:	Up to 200°C
Pressure:	Up to 15 bar (absolute). In case of a pressure >2 bar, the maximum temperature is limited to 30°C
Connectors:	Swagelok for 6 mm outer diameter tubing

4.7.2 DryPath™

The DryPath™ option allows to reduce the concentration levels of water and carbon dioxide in the optics compartment of the FTIR spectrometer without the need for an external purge gas supply. This option is particularly suited for the quantification of water and carbon dioxide in low concentrations much below ambient conditions.

With the DryPath™ option, purging OMEGA 5 is not required. If the DryPath™ option is ordered, the option is installed during production of the OMEGA 5 gas analyzer. Additional installation steps on site are not required.

Whenever the OMEGA 5 gas analyzer is switched on, the DryPath™ option is put to use, i.e. the optics compartment of the FTIR spectrometer is purged automatically.

Technical details on the DryPath™ option are listed in appendix A.10. Details on the maintenance of the DryPath™ option are described in chapter 6.4.

5 First steps

If the gas analyzer has been configured and connected to the measurement PC, the gas analyzer is ready to operate. The gas analyzer is completely software controlled, i.e. many components (e.g. gas cell unit) are selected and controlled by the software.

This chapter describes the basic operation of the gas analyzer using the OPUS GA spectrometer software. Further details on the OPUS GA software are described in the OPUS GA user manual.

5.1 Operating safety



CAUTION

Objects placed on the gas analyzer

Risk of personal injury due to objects which may fall down.

- Do NOT place any object on top of the gas analyzer.

5.2 Purging the gas analyzer

If you do not use the DryPath™ option (chapter 4.7.2), it is recommended to purge the gas analyzer by purge gas to lower the levels of humidity and carbon dioxide in the optics compartment.

5.2.1 Safety



WARNING

Inflammable gases in connection with the gas analyzer

Serious injury or death

- NEVER use inflammable gases in connection with the gas analyzer.

WARNING



Using dry nitrogen to purge the gas analyzer

Risk of asphyxiation

- Only work in well-ventilated rooms.
- If required, install oxygen warning devices as excessive nitrogen gas concentrations in the air reduce the concentration of oxygen. As nitrogen gas is colorless, tasteless, and odorless there is the risk of being inhaled as if it were normal air.

CAUTION



Highly toxic, corrosive and/or explosive liquids in connection with the gas analyzer

Risk of personal injury

- NEVER use highly toxic, corrosive and/or explosive liquids in connection with the gas analyzer.
- Highly toxic, corrosive and/or explosive liquids in connection with the gas analyzer may damage the gas cell windows due to overpressure or corrosion. The explosive gas mixture may enter the optical compartment of the spectrometer and inflame the IR source.

-  Do not fill liquid samples into the gas analyzer. Only apply the vapor phase of the sample.
-  Check the gas flow by using a flow meter to make sure that the pressure inside the gas cell does not decrease due to damage of the gas cell or gas cell windows.

5.2.2 Purge gas requirements

Purge gas:	• CO₂-reduced and dry air • dry nitrogen
Maximum pressure:	0.5 bar
Gas rate:	5 l/minute at maximum ^a
When must be purged?	If the ambient air humidity level inside the gas analyzer is too high. ^b

a. You can control the purge gas flow rate by means of a flow-rate meter.

- b. The purging procedure reduces the level of water vapor, CO₂ or other contaminants inside the gas analyzer. Water vapor, CO₂ and other atmospheric contaminants cause unwanted peaks in the mid-infrared spectrum range.

5.3 Switching on gas analyzer

i When the gas analyzer was stored outside the operating temperature range specified, the gas analyzer must have been acclimatized before you can switch it on.

1. If not already done, connect the gas analyzer to the power supply (chapter 3.6.1), and to the PC (chapter 3.6.4).
2. At the gas analyzer rear side, turn the main power switch into the "I" (ON) position.
3. To increase/decrease the temperature for the integrated transfer line or gas cell,

press the  and/or  button.

CAUTION



No protective thermal hose installed around the fittings connected to the integrated transfer line

Risk of skin burn

- If you quantify gases at hot temperatures (>37°C), always install a protective thermal hose around the fittings connected to the transfer line.
- Do not touch the rear housing of the gas analyzer during ongoing operation.

4. Inspect the measured temperature value on the particular temperature controller to make sure that the gas analyzer and the possibly attached transfer tube reach the appropriate operating temperature.
 - *A measurement is only reasonable after the operating temperature has been reached (compare target and actual value).*

i After powering up the gas analyzer wait at least 10 minutes before starting with the first measurement. Advanced quantitative measurements require an even longer period of stabilization time.

5.4 Starting software

1. Switch on the measurement PC.
2. In the C:\OPUS GA directory, double click the *OPUS GA.exe* file.
3. Wait until the *Times Series* window of OPUS GA is shown.
 - Numerous calculations are performed to load existing reference spectra. Depending on the number of reference spectra to be loaded, the start-up procedure may take some time.
 - The gas compounds which can be identified in the spectrum are shown in the left table (figure 5.1). Only those gas compounds selected by a colored box will be evaluated when new spectra are recorded.

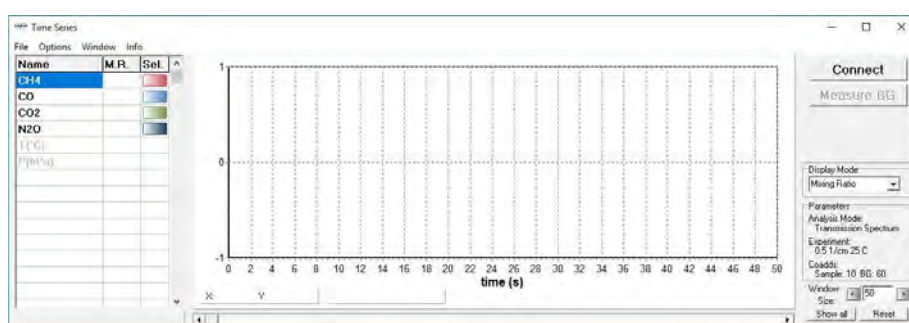


Figure 5.1: OPUS GA - *Times Series* window

4. From the left table (figure 5.1), select the gas compounds (30 at maximum) to be evaluated.
 - Gas compounds are selected/deselected by clicking the cells in the 'Sel.' column. The colored box indicates that the cell is selected.
 - When right-clicking a selected cell, the box color becomes white, indicating that the particular gas compound will be evaluated but not visualized during measurement.
5. Click the *Connect* button to establish the connection between PC and gas analyzer.
 - During connection, all system components are initialized sequentially. The current status is shown in the status bar of the start view.
6. Wait until the connection has been established.
 - The connection between PC and gas analyzer has been established if the 'Connect' button has changed to 'Start'.
 - Since the detector is attached inversely to the analog/digital converter (ADC), higher intensities cause a decrease of the digital signal. Therefore, the maximum value of the interferogram is negative. The signal always should be well within the range -1 to 1, as this corresponds to the maximum range of the ADC.

5.5 Setting number of scans to be averaged

1. On the *Options* menu of the *Time Series* window, select the *Preferences* command or press the *F4* button.
2. On the dialog that opens, set the number of scans to be averaged.
➤ *Details on the setting options are described in chapter 5.5.1.*
3. Click OK to confirm.

5.5.1 Details on setting options

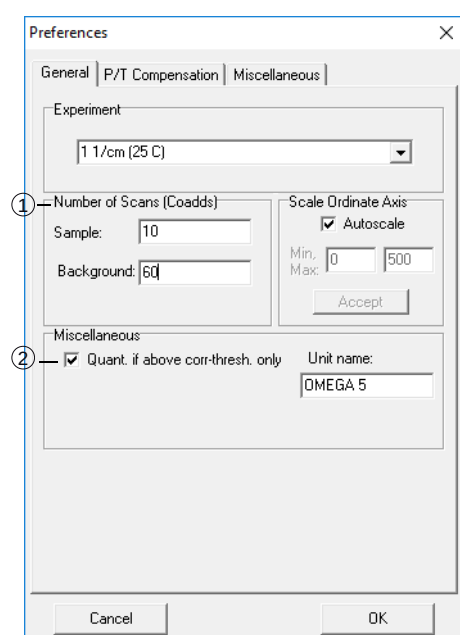


Figure 5.2: Entering the number of scans to be averaged

	Component	Definition
1	Number of Scans (Coadds)	The number entered defines the number of scans which will be averaged for the sample and background measurement ^a .
2	Quant. if above corr-thresh. only	If activated, retrieved gas concentrations will not be shown if the spectral signature was not unambiguously enough in the recorded spectrum.

Table 5.1: Definition of setting options

- a. A background measurement is a measurement without any sample available in the gas analyzer. The purpose of the background measurement is to see how the ambient conditions (level of air humidity, temperature etc.) and the gas analyzer itself can influence the calculation of the sample spectrum. The measured background spectrum is used to calculate the transmission spectrum.

5.6 Starting measurement

1. Purge the gas cell using reference gas.
 - *The reference gas should be as free as possible from infrared-active gas compounds. Gaseous nitrogen or dried carbon dioxide reduced air are recommended. Details on the purge gas requirements are described in appendix A.8.*

WARNING



Using dry nitrogen to purge the gas analyzer

Risk of asphyxiation

- Only work in well-ventilated rooms.
- If required, install oxygen warning devices as excessive nitrogen gas concentrations in the air reduce the concentration of oxygen. As nitrogen gas is colorless, tasteless, and odorless there is the risk of being inhaled as if it were normal air.

2. In the *Time Series* window, click the *Measure BG* button to start the background measurement.
3. Purge the gas cell using the analysis gas.
4. In the *Time Series* window, click the *Start* button.
5. On the *Save Scan Data* dialog (figure 5.3), specify the path to be used to save the measurements. Either type the path name manually or click the *Choose Folder* button.

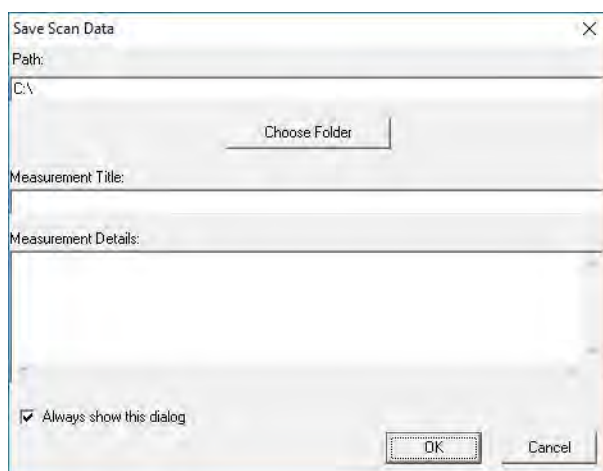


Figure 5.3: Save scan data

- *For each measurement a separate sub-folder will be created. By default, the sub-folder name contains the 'yyyy_mm_dd_HH_MM_SS'¹ prefix, indicating the date and time of the measurement.*

1. YYYY: year, MM: month, DD: day, HH: hour, MM: minute, SS: second

6. Define a measurement title, which will be the suffix of the sub-folder name.
 ➤ *If no specific measurement title has been defined, the default title will be 'Measurement'.*
7. If required, add measurement details which specify the particular measurement.
 ➤ *The measurement details are saved in an info file within the sub-folder.*
8. Click OK to confirm.
 ➤ *The measurement starts. Clicking the 'Stop' button completes and saves the current measurement.*

5.6.1 After measurement has finished

After the measurement has finished all data are saved in the path defined. The following data categories are saved:

Data category	File extension	Format
Mixing ratios of gas compounds	*_ResultSeries.txt	ASCII
Sample interferogram	*.ifs	Binary
Sample spectra	*.ssp	Binary
Measurement details	*_Info-File.txt	ASCII
Temperature/pressure log	*_PT_Log.txt	ASCII
Time information	*.tim/*_abt	ASCII
Background information	*_background.ifg	ASCII
Background spectrum	*_background.prn	ASCII
System files	*.prn/*_ini	Plain text/INI format
Spectrometer report	*.htm	HTML

Table 5.2: Extensions and formats of measurement results

5.7 Viewing and analyzing measurement results

During measurement, the gas compounds previously selected (chapter 5.4) are quantified online. The central panel of the *Time Series* window shows the concentration as a function of the time, by default in ppm (figure 5.4).

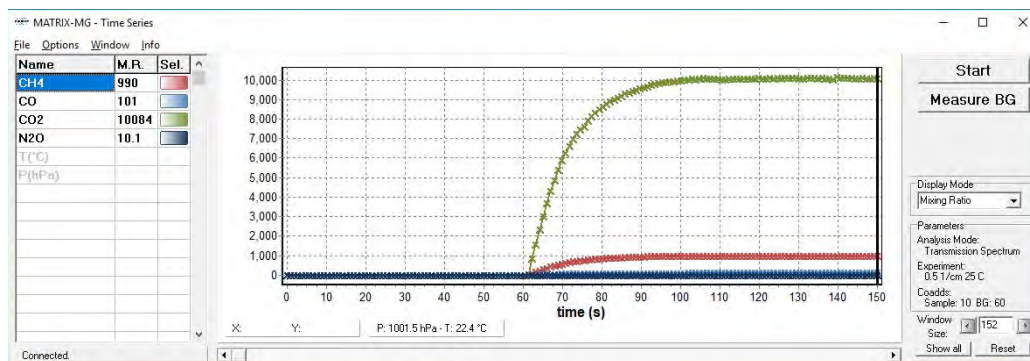


Figure 5.4: Measurement result of the gas compounds selected

Up to 30 gas compounds can be selected simultaneously. Gas compounds which are not selected will not be evaluated, and their concentration values are not shown.

If you focus on a particular gas compound at post-evaluating, which was not selected before, select this particular gas compound and reload the measurement results (chapter 5.8).

To analyze the measured spectra in further detail, select the *Spectral Analysis* command on the *Window* menu of the *Time Series* view.

The *Spectral Analysis* view shows the measured spectra and contains various views to analyze the model fit to the measured spectra.

5.8 Loading measurement results

- On the *File* menu, select the *Load/Reanalyse Spectrum/Interferogram Series* command.
 - In this context, the term 're-analyse' means that the fits to each spectrum are evaluated once again.
- On the dialog that opens, select the particular interferogram (*.ifs) or spectrum (*.ssp) series.
 - If an interferogram series is selected, the spectra will be calculated again from the measured interferograms using the currently defined Fourier transform parameters.
 - If a spectrum series is selected, the spectra already calculated from the interferograms during the measurement will be opened.
- Click OK to confirm.

6 Maintenance

The gas analyzer is a low-maintenance instrument, i.e. the operator can replace components with a limited service life (e.g. fuses). Any kind of maintenance work not described in this chapter is only allowed to be done by Bruker Service.

- Replacing fuses
- Replacing defective power cord
- Replacing DryPath™ filter
- Cleaning

6.1 General maintenance considerations

Perform only the maintenance procedures described in this chapter. Strictly observe the relevant safety precautions. Any failure to do so may cause property damage or personal injury. In this particular case Bruker does not assume any liability.

Maintenance procedures not described in this manual should only be performed by a Bruker service engineer. To ensure user and property safety, disconnect the power supply before performing any maintenance procedures.

6.1.1 Electrostatic discharge

Electronic components (e.g. semiconductor chips, printed circuit boards) are very susceptible to electrostatic discharges. Even discharge which may not be perceptible to the operator, and which the operator causes by touching the electrical ports, can damage electronic components inside the gas analyzer.

To avoid uncontrolled electrostatic discharging caused by the operator, the operator should touch an already grounded object (e.g. radiator) or use a grounding wrist strap.

6.1.2 Disposal

- Observe national and locally applicable directives as well as legal provisions pertaining to the safe disposal
- Recycle the material separately
- Ensure an environmentally sound disposal of all component parts according to the legal provisions

6.2 Replacing fuses

i When replacing the fuses only use one of the following fuse types:

- 2 x T 8 AH, in case of 110 V
- 2 x T 4 AH, in case of 230 V

according to IEC 60127-2. The original spare parts can be ordered at Bruker Optik GmbH (appendix B.4). **Warning:** A gas analyzer set to 100 V to 120 V at the factory must not be operated at higher voltages as the fixed heating resistances cause excessive currents and temperatures. The heating system can be damaged by too high currents and temperatures. Changing the operating voltage from 120 V to 230 V or vice versa can be done by Bruker Service.

The fuse holder (blew marking in figure 6.1) is located at the gas analyzer rear side below the I/O switch (main power switch).



Figure 6.1: OMEGA 5 - Fuse holder

1. Put the I/O switch, located on the gas analyzer rear side, in the "O" (OFF) position.
2. Unplug the power cord from the power supply and gas analyzer.

3. Open the fuse holder flap.

➤ *Auxiliary tool: Insert a small flat-ended screwdriver into the slot and gently raise the screwdriver to release the spring clip of the fuse holder (figure 6.2).*

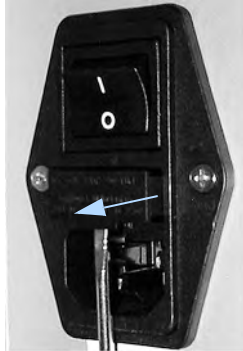


Figure 6.2: Opening fuse holder

4. Open the fuse holder.
5. Replace both fuses (figure 6.3), even if only one fuse is defective.



Figure 6.3: OMEGA 5 - Fuses

6. Press the flap against the housing until the spring clip engages.
7. Reconnect the power cord to the power supply and gas analyzer.
8. Put the I/O switch, located on the gas analyzer, in the "I" (ON) position.

6.3 Replacing defective power cord

Replace the power cord if any damage to the insulation, connectors or cable is visible. Never repair a damaged power cord, but replace it by an intact power cord rated for at least 230 V, 4.5 A.

Only use an original Bruker power cord. The original spare part can be ordered with Bruker (appendix B).

6.4 Replacing DryPath™ filter

If the gas analyzer is used in connection with the DryPath™ option, the DryPath™ filter must be replaced on a regular basis to ensure a constantly low level of water vapor and carbon dioxide in the optics compartment of the spectrometer.

The typical lifetime of a DryPath™ filter is about 1 year, but depends on the operating time and ambient conditions (humidity, temperature, pressure fluctuations, volume fraction of CO₂). A necessary filter change is indicated by a strong increase of the CO₂ background volume fraction (>10 ppm).

1. Put the I/O switch, located on the gas analyzer rear side, in the "O" (OFF) position.
2. On the gas analyzer upper right rear side, carefully pull out the used filter from the housing (figure 6.4).

➤ *The filter protrudes from the housing by 3 cm, which provides easy grip. As the filter is firmly attached to its mounting position you may apply some force when pulling it out.*

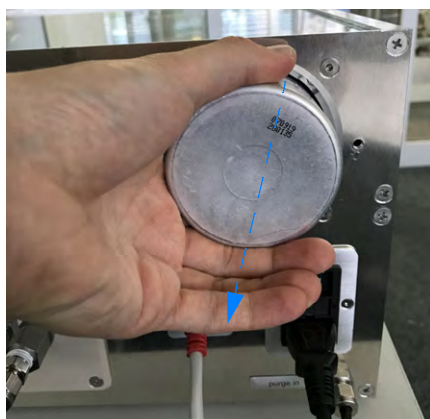


Figure 6.4: Pulling out DryPath™ filter

3. Remove the packaging film from the new filter¹ (figure 6.5).
4. Pull off the filter cap and remove it.



Figure 6.5: DryPath™ filter

1. The replacement filter can be ordered with Bruker. The spare part number is indicated in appendix B.1.

5. Carefully insert (figure 6.6) the new filter into the designated mounting at the rear gas analyzer side until it audibly snaps into its position.



Figure 6.6: Inserting DryPath™ filter

6. Put the I/O switch, located on the gas analyzer rear side, in the "I" (ON) position.

6.5 Cleaning

The outer gas analyzer housing can be cleaned only by using a dry or damp cloth. Do not use detergents with organic solvents, acid or base!

7 Troubleshooting

This chapter describes possible gas analyzer problems, their potential causes and recommended solutions.

Not all failures and causes can be outlined in this chapter. If the recommended solutions described in this user manual do not solve the problem, contact Bruker Service (chapter 1.4).

7.1 Hardware problem

Problem	Troubleshooting
• The gas analyzer is not operating at all. • The internal fan is not working.	• Check the mains supply to the gas analyzer. • Check whether the gas analyzer is switched on. • Check the main fuses^a mounted inside the on/off switch, located at the gas analyzer rear side. • If the power supply unit is defective, contact Bruker Service (chapter 1.4).
Continuous beeping after switching on gas analyzer.	Check whether the Ethernet cable is firmly connected to the gas analyzer and/or PC.

Table 7.1: Hardware problems

a. In case of 110 V: 2 x T 8 AH; in case of 230 V: 2 x T 4 AH

7.2 Analysis problems

Problem	Troubleshooting
Measurement error in OPUS GA during measurement	• Run the system <i>SysDiag</i> diagnosis tools or download the <i>SysDiag</i> file using the following link: https://storage.driveonweb.de/dowdoc/214b699a29988cc6d15d175116ce9c073e23752c3b9cd9e7 • Unzip the file in the C:\OPUS GA\ folder. • Start OPUS GA. • Select <i>Info > Collect System Data</i>. <i>➤ The data will be saved in the 'C:\OPUS GA\SysDiag\' path.</i> • Copy the latest diagnosis zip file into the C:\OPUS GA\SysDiag\ folder and send it to Bruker Service (chapter 1.4).
No connection between OPUS GA and gas analyzer	• Run the <i>SysDiag</i> system diagnosis tool as described above. • Generate a Full Report as described in chapter 8.2. • Send the zipped diagnosis file from OPUS GA, and the Full Report to Bruker Service (chapter 1.4).
Overall or partial spectrum intensity decreases significantly	Run a measurement in OPUS GA and send the resulting measurement folder to Bruker Service (chapter 1.4). Possible malfunctions are: • rough or dirty gas cell mirrors or windows due to the condensation of corrosive or sticky gas compounds • declining intensity of IR source • declining sensitivity of thermoelectrically cooled MCT detector

Table 7.2: Analysis problems

Problem	Troubleshooting
Strong water and CO ₂ lines in the background spectrum even after hours of purging the gas cell with dry CO ₂ -reduced gas or pure nitrogen	If the DryPath™ option is used: • Make sure that the gas cell is purged to be free of traces of water and carbon dioxide. • Exchange the used DryPath™ filter (chapter 6.4). The spare part number of the replacement filter is indicated in appendix B.1. If the DryPath™ option is not used: • Install a purge gas connection (chapter 3.6.2) to purge the optics compartment of the spectrometer. • Use pure nitrogen or dry CO₂-reduced air to purge the optics compartment. The purge gas requirements are described in appendix A.8.

Table 7.2: Analysis problems

8 Web Interface

8.1 General information on the firmware pages of the gas analyzer

Due to the integrated EWS15 board (Embedded Webserver Board), the gas analyzer operates like a Web server, i.e. the gas analyzer can be accessed using a Web browser (e.g. Microsoft Internet Explorer, Netscape Internet Browser). This kind of gas analyzer accessing requires a physical connection between the gas analyzer and PC as well as knowledge about the gas analyzer IP address.

If you open the Web browser and enter the gas analyzer IP address (default IP address: 10.10.0.1) into the address field of the Web browser (① in figure 8.1), the firmware start page of the particular gas analyzer is shown. If you do not know the IP address, see the white label at the electronics panel at the gas analyzer rear side.



Figure 8.1: Firmware start page - OMEGA 5

8.1.1 User authentication

To ensure network reliability the gas analyzer firmware provides a user authentication via Web browser for sensitive instrument configuration pages, and blocks unauthorized modification of the instrument configuration. The protected configuration pages are:

- *Measurement Menus*
- *Direct Command Entry*
- *Service/List of result data files*
- *Service/Special commands A*
- *Service/Special commands B*
- *Service/Reset Instrument*
- and all *Service/Edit hardware configuration* pages, except for the *Backup Flange Eeprom data to Flashdisk* and *Backup NVRam to Flashdisk* pages

The user authentication has no influence on the communication connection established to the gas analyzer via OPUS.

i Only one user account (combination of user name and password) is supported. A particular user can be logged on from several workstations at the same time.

8.1.1.1 Activating user authentication

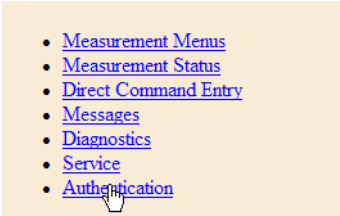
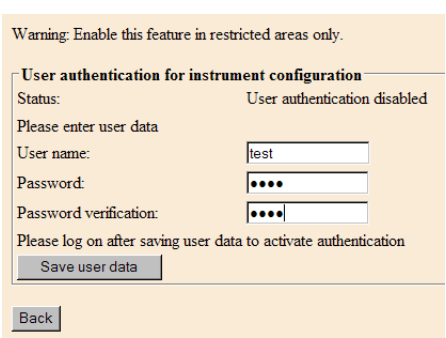
1		• Open the Web browser. • Enter the gas analyzer IP address into the address entry field. • On the configuration page, click <i>Authentication</i>.
2		• Enter the user name and password. • Confirm the password.

Table 8.1: Activating user authentication

3	Warning: Enable this feature in restricted areas only. User authentication for instrument configuration Status: User authentication disabled Please enter user data User name: test Password: Password verification: Please log on after saving user data to activate authentication Save user data	Click the <i>Save user data</i> button.
4	Warning: Enable this feature in restricted areas only. User authentication for instrument configuration Status: Not logged out or user data ("test") modified - please log on Log on Back	Click the <i>Log on</i> button.
5		• Enter the user name and password. • Click <i>OK</i> to confirm.
6	Warning: Enable this feature in restricted areas only. User authentication for instrument configuration Status: user "test" logged on Log off Disable authentication: Disable Please enter user data: New user name: test New password: Password verification: Save user data Back	The status line now reads: <i>user "xxx" logged on.</i> i To have the user authentication become effective you have to log off.
7	Warning: Enable this feature in restricted areas only. User authentication for instrument configuration Status: user "testa" logged on Log off Disable authentication: Disable	Click the <i>Log off</i> button.

Table 8.1: Activating user authentication

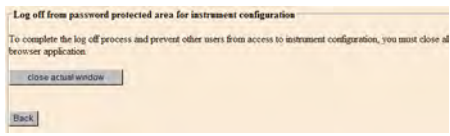
8		• Click the <i>close actual window</i> button. • Click <i>Yes</i> to confirm the action.
9	• Open the Web browser. • Enter the gas analyzer IP address again. i The access to the instrument configuration pages is not possible any more. As soon as you try to open one of these pages a dialog pops up, asking for the user name and password. If you have entered the user name and password, you will be allowed to access the protected instrument configuration pages again.	

Table 8.1: Activating user authentication

8.1.1.2 Deactivating user authentication

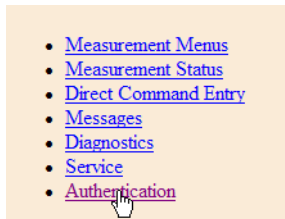
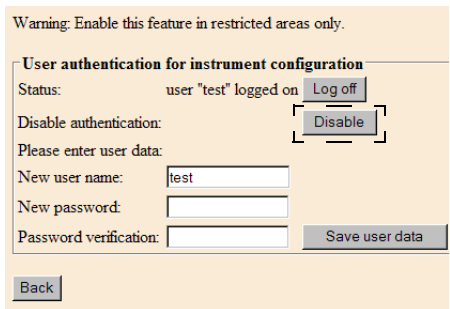
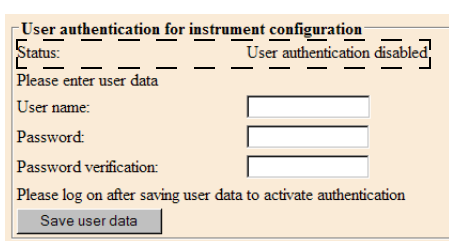
1		• Open the Web browser. • Enter the gas analyzer IP address into the address entry field. • On the configuration page, click <i>Authentication</i>.
2		Click the <i>Disable</i> button.
3		The status line now reads: <i>User authentication disabled</i> .
4		Click the <i>Back</i> button to return to the configuration page again.

Table 8.2: Deactivating user authentication

8.2 Full report

The full report is a useful fault diagnostics tool. This report provides all the necessary information about the instrument configuration and complete gas analyzer status.

This report also includes error messages, internal and external data transfer as well as values of all relevant parameters. The full report is an important and - especially if a problem or failure has occurred - indispensable tool for fault diagnostics by service engineers.

8.2.1 Generating full report

1	• Measurement Menus • Measurement Status • Direct Command Entry • Messages • Diagnostics • Service • Authentication	• Open the Web browser. • Enter the gas analyzer IP address into the address entry field. • On the configuration page, click <i>Service</i>.
2	• << One Level Back • View Instrument Configuration • Full Report • Log buffer • Beep • List of commands • List of recent data files	Click the <i>Full Report</i> option. The current full report is displayed.
3	• On the <i>File</i> menu of the Web browser, click the <i>Save as</i> command. • Use the file extension <i>*.htm</i>. • Send the original htm file to Bruker (service.bopt.us@bruker.com). Do not generate any screenshots or text files.	

Table 8.3: Generating full report

i Save the full report immediately after a problem or failure has occurred. Otherwise, important information will be overwritten by newer entries.

A Specifications

A.1 General

Parameter	Specification
Dimension	Gas analyzer: 740 x 485 x 222 mm (d x w x h)
Regulations	• Complies with IEC/EN 61010-1 (Safety requirements for electrical equipment for measurement, control and laboratory use) • Complies with IEC/EN 61326-1 (Electrical equipment for measurement, control and laboratory use - EMC requirements)
Weight	Gas analyzer: 28 to 29 kg (depending on configuration)
Power consumption (basic configuration)	• minimum: 30 W when heater is off • typical: 85 W in average, when the heaters reached the target temperature of 190°C (with ambient temperature: 25°C) • maximum: 1100 W at the start of heating
Power cord	3 x 0.75 mm ² with safety plug, according to VDE 0620-1 or IEC
Overvoltage category ^a	II, complies with 61010-1 or IEC 60664-1
Power supply	• Voltage: 110 V ±10%/60 Hz or 230 V ±10%/50 Hz • Protection class: I, according to IEC 61140

Table A.1: Specifications - General

a. The values indicated are target values which have an effect on the performance of the entire configuration. In case of non-observance the functioning and safety of the configuration may be affected adversely.

A.2 Labeling on gas analyzer

Parameter	Specification
Labeling on gas analyzer	①  ② Bruker OPTIK GmbH Rudolf-Plank-Str. 27 D-76275 Ettlingen ③ WEEE-Reg-Nr. DE84716930 ④ Type: OMEGA 5 ⑤ Ser.No.: 0001 ⑥ ECL: 00 ⑦ Prod.date: 01/2020 ⑧ 110V; 8A; 60Hz ⑨ Sch./Fuse - IEC 60127-2 2x T 8A / 250V → ⑩  • 230 V: ①  ② Bruker OPTIK GmbH Rudolf-Plank-Str. 27 D-76275 Ettlingen ③ WEEE-Reg-Nr. DE84716930 ④ Type: OMEGA 5 ⑤ Ser.No.: 0001 ⑥ ECL: 00 ⑦ Prod.date: 01/2020 ⑧ 230V; 4,5A; 50Hz ⑨ Sch./Fuse - IEC 60127-2 2x T 4A / 250V → ⑩ 

Definition

1

Logo of manufacturer

2

Company address of manufacturer

3

Registrations number which ensures the take-back of waste by the manufacturer, in compliance with the WEEE directive.

4

Instrument type

5

Instrument serial number

6

Instrument engineering change level

7

Production date

8

Details on power supply and power consumption

Table A.4: Specifications - Labeling on gas analyzer

Parameter	Specification
	Definition
9	Fuse types The arrow indicates the position of the fuse at the gas analyzer.
10	- Conformity marking for the European (CE) economic market - The warning symbol indicates that the user should observe the instructions given in the user manual before starting installation, operation, repair and/or maintenance of the instrument.

Table A.4: Specifications - Labeling on gas analyzer

A.3 Performance

Parameter	Specification
Spectral range	4,500 - 800 cm^{-1}
Spectral resolution	Better than 2 cm^{-1} ; optional: better than 1.0 cm^{-1}

Table A.5: Specifications - Performance

A.4 Optics

Parameter	Specification
Detector	Thermoelectrically cooled MCT detector
Drying	• By means of a purge gas • DryPath™ option: drying is possible without a purge gas supply required
Interferometer	RockSolid
Source	CenterGlow IR source

Table A.6: Specifications - Optics

A.5 Electronics

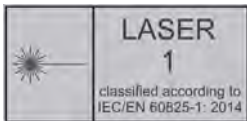
Parameter	Specification
A/D converter	100 kHz with 24 bit dynamic range
Fuse	• 110 V: 2 x T 8 AH 250 V, according to IEC 60127-2 • 230 V: 2 x T 4 AH 250 V, according to IEC 60127-2
Laser class	OMEGA 5 is a product of laser class 1.^a The following label identifies OMEGA 5 as a product of laser class 1:  The label is located at the rear side of the gas analyzer.
Laser diode	850 nm
Ports	• ETH/LAN • Power
Sensors	Integrated temperature and pressure sensors

Table A.7: Specifications - Electronics

a. If the gas analyzer is used as specified for intended use (chapter 1.3.3), laser radiation is not accessible for the user.

A.6 Environment

Parameter	Specification
Atmospheric pressure	80 to 105 kPa
Installation site	In a closed room, up to a maximum height of 2000 m above sea level
Operational temperature range	5°C to 35°C
Storage and transport temperature	-20°C to 70°C
Humidity (non-condensing)	Less than 80% (relative humidity)
Temperature variation	Max. 1°C/ per hour and max. 2°C per day
Pollution degree	2, complies with 61010-1 or IEC 60664-1

Table A.8: Specifications - Environment^a

a. The values indicated are target values which have an effect on the performance of the entire configuration. In case of non-observance the functioning and safety of the configuration may be affected adversely.

A.7 Gas cell

Parameter	Specification
Type	Multi-reflection cell
Optical path length	5 m
Volume	about 330 ml
Cell body	Aluminium, nickel-coated
Window material	Zinc selenide
Mirrors	Gold-coated
Maximum temperature	191°C
Maximum pressure	2 bar absolute
Maximum gas flow	10 NI/min

Table A.9: Specifications - Gas cell

A.8 Purge gas

Parameter	Specification
Purge gas properties	• Air or nitrogen gas • Dry and CO₂-reduced: dew point below -40°C, which corresponds to a degree of dryness of 128 ppm, CO₂ <0.1 ppm • Dust/mist and oil-free: depending on the particle load and size, the purge gas should be preprocessed by a dust/particulate filter connected to a fine filter with a pore size of 0.5 to 7 µm, as the circumstances require
Temperature	Ambient temperature
Pressure	• Standard: 50 mbar (0.725 psi) overpressure • Maximum: 0.5 bar (7.25 psi) overpressure
Flow rate	The optimum flow rate depends on the specific application. • Standard: 1 NI/min • Maximum: 5 NI/min

Table A.10: Specifications - Purge gas

A.9 Sample gas

Parameter	Specification
Sample gas properties	• Chemically and thermally stable • Non-condensing: dew point well below the temperature of the gas cell and of the inlet and outlet. Note that the coldest parts of the gas cell, e.g. the windows, can be several degrees colder than the set gas cell temperature. • Dust and oil free: depending on the particle load and size, the sample gas should be preprocessed by a dust/particulate filter connected to a fine filter with a pore size of 0.5 to 7 μm, as the circumstances require • Mist free: a mist filter or cooler should be installed to avoid condensation of the gas compound in the gas cell • Non-flammable, non-explosive • Non-corrosive: volume mixing ratio of a corrosive substance (such as F_2, Cl_2, HF, HCl, H_2SO_4, ClF_3, O_3) should not exceed 10,000 ppm, in particular in case of water vapor at volume mixing ratios in the range of 1,000 ppm or higher
Temperature	Up to 191°C
Pressure	• Up to 2 bar (29 psi) absolute • In case of OMEGA 5-HP high-pressure option: up to 15 bar (217.6 psi) absolute
Flow rate	The optimum flow rate depends on the specific application. • Standard: 1 NI/min • Maximum: 10 NI/min

Table A.11: Specifications - Purge gas

A.10 DryPath™ option

Parameter	Specification
Purpose	To measure water vapor and carbon dioxide in low concentration much below ambient conditions (4 to 5 orders of magnitude). No external purge gas supply is required.
The typical effective background contribution of the optics compartment is	• <10 ppm for water • <1 ppm for CO₂
Diurnal stability of measured value for zero air	• <1 ppm for water • <0.1 ppm for CO₂ Both values at 50% relative humidity, 25°C, 1013 bar
Expected lifetime	1 year

Table A.12: Specifications - DryPath™

B Spare Parts and Accessories

B.1 Consumables

Part No.	Description
1873072	Thermoelectrically cooled MCT detector
Q328/OMEGA	MIR source, pre-mounted, pre-aligned
S121/D	Replacement filter set for DryPath™ option

Table B.1: Consumables

B.2 Gas cell spare parts

Part No.	Description
1822789	ZnSe window (wedge 6 mrad, 19 x 4 mm, BBAR coating)
1827649	Zone mirror
1834533	Pressure transmitter; 0 to 2 bar, 4 to 20 mA
1834537	Pressure transmitter (OMEGA 5-HP); 0 to 2 bar, 4 to 20 mA
1834541	Surface resistance thermometer PT100
1836792	Object mirror
F140-17	BaF ₂ window (wedge 6 mrad, 19 x 4 mm)
F140-3	CaF ₂ window (wedge 6 mrad, 19 x 4 mm)

Table B.2: Gas cell spare parts

B.3 Parts with unlimited lifetime

Part No.	Description
1008354	Power cord, 3X0.75QMM/1.8M SW C5-USA
1012365	Power cord, 1.8M SW C5-CHINA IEC6032
1828376	Power cord, 3X0.75QMM/2M SW C5-EURO-90°

Table B.3: Parts with unlimited lifetime

B.4 Accessories

Part No.	Description
1801526	FUSE MINI 5X20MM 4.0A T
1801423	FUSE MINI 5X20MM 8.0A T
I22406	Tool kit

Table B.4: Accessories

B.5 Software

Part No.	Description
B-GA-HR	FT-IR spectra library for OPUS GA
B-GA-HR_EXT	FT-IR spectra library for OPUS GA with additional high-resolution gas phase spectra
O/GA	Gas analysis software OPUS GA
O/GA_IND_VERS	OPUS GA, upgrade to individual version
O/GA-NLU	OPUS GA, additional license
O/GAINDVERS-NLU	OPUS GA, additional license, upgrade to individual version

Table B.5: Software

C Dimensional Drawings

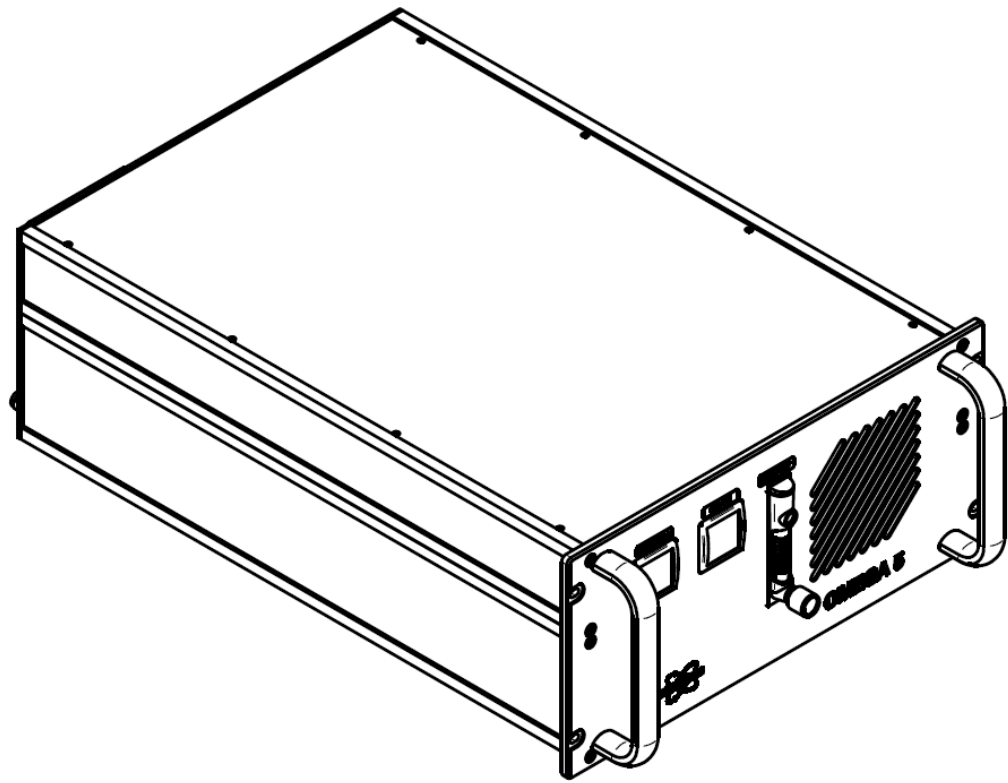


Figure C.1: OMEGA 5 - Isometric view

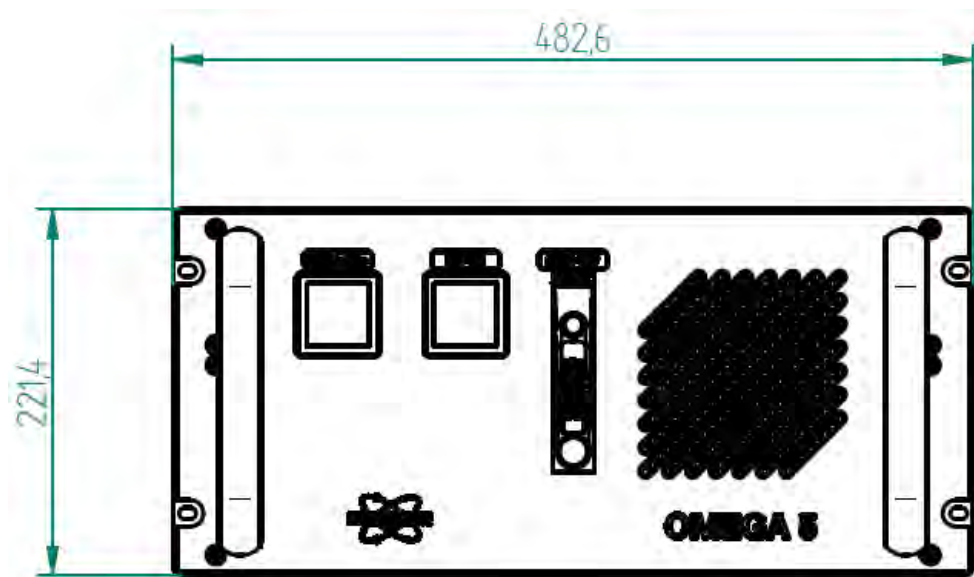


Figure C.2: OMEGA 5 - Front view with (dimensions in mm)

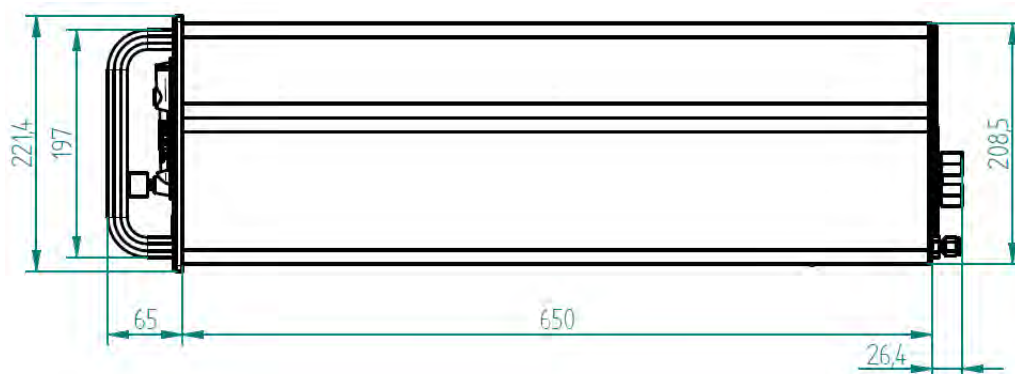


Figure C.3: OMEGA 5 - Right side view (dimensions in mm)



Figure C.4: OMEGA 5 - Left side view (dimensions in mm)

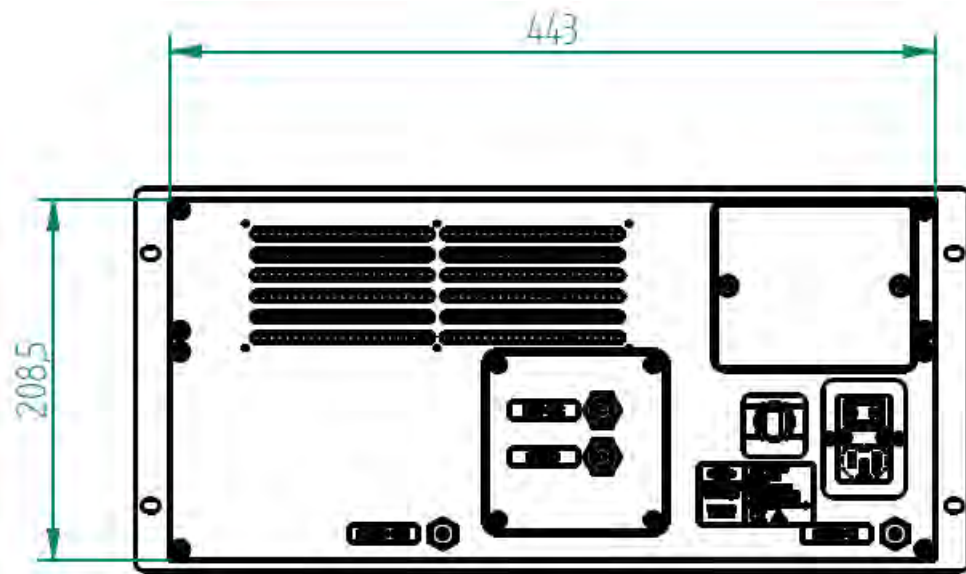


Figure C.5: OMEGA 5 - Rear view (dimensions in mm)

D Glossary

Detector	A detector converts incoming light into an electrical signal.
Ethernet	Ethernet is the most widely installed Local Area Network (LAN). A LAN is a network of interconnected workstations sharing the resources of a single processor or server within a relatively small geographic area (i.e. an office building). Ethernet is a set of hardware and signaling standards used for LANs. The most commonly installed systems are 10/100/1000 BaseT.
FTIR	The <u>F</u>ourier <u>T</u>ransform <u>I</u>nfra<u>R</u>ed Spectroscopy is a method to obtain infrared spectra of a sample using an interferometer. A Fourier transform is performed on the resulting interferogram to calculate the spectrum. In case of an FTIR spectrometer infrared light emitted from a source is directed to an interferometer which modulates the light. Leaving the interferometer the light passes through the sample compartment (and also the sample) and is focused onto the detector. The signal measured by the detector is called the interferogram.
Interferometer	An interferometer detects interferences, that means overlapings of light waves. Inside the interferometer a light beam is split into two light waves (beams) by the beam splitter. These two light waves pass through optical paths of different lengths, are reflected by additional mirrors, and finally recombined. If you change the optical path of one of the two light waves, e.g. by moving one mirror, the phases of the two light waves displace against each other. When recombining the light waves, interferences occur. The result is an interference pattern (stripes or rings) which has been written by the light waves. This pattern is determined by the difference of the optical path lengths which the single light waves have passed before being recombined.
IP address	An IP address is the network address of a workstation or network. IP addresses consist of 4 number fields separated by dots. Each number field represents 1 byte. Values can be between 0 and 255. The numbers on the left of the string define the network, the numbers on the right define the individual workstation or Network Interface Card (NIC).
Relative humidity	Relative humidity is the amount of water vapor in the air, which can be between 0 and 100%.

Sample	A sample is a special kind of substance which is put into the gas analyzer sample position, and measured.
Signal-to-noise-ratio	The signal-to-noise ratio describes the quality of a wanted signal which has been superimposed by noise. It outlines the ratio between the average performance of a wanted signal emitted by a signal source, and the average performance of noise.
USB	The <u>U</u> niversal <u>S</u> erial <u>B</u> us (USB) is a serial bus system which allows to connect peripherals to a host computer. Devices or storage media equipped with USB can be hot plugged, and the properties of peripherals can be automatically detected.

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